

Chapter 5

Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs

5.1 Introduction and Scope

Chapters 2 and 4 established and quantified the slow, volatile response of these polymer-electrolyte composites. Behavioural models fitted to that response are now evaluated on temporal-computing tasks. A reservoir maps a time-varying input onto a high-dimensional dynamical state from which a linear read-out can recover temporal information.

The central comparison is between a bank drawn from one composition and a bank spanning several measured timescales. The latter should help when a target depends on several lags, but need not improve a task dominated by one timescale. Wearable *affective computing* (the recognition of emotional and physiological state) provides a relevant

test because its signals contain both fast and slow components. The evaluation proceeds from reservoir benchmarks to physiological streams, WESAD classification, and tests with sensor noise, an independent corpus, and wrist-band data.

All results in this chapter are *in silico*; no new devices were fabricated. Each node is a compact behavioural model parameterised by the fits in chapter 4 (section 5.2), and every read-out is a single linear ridge regression. Demonstration A tests whether one composition suffices for a single-timescale feature. Demonstration B tests whether a composition-diverse bank improves multi-timescale tasks.

5.2 Measured Behavioural Model

Each composition cell is reduced to a compact behavioural model and used as one reservoir node. The model is fitted only to the three common measurements of chapter 4: hysteresis, pulsed potentiation, and delayed depotentiation. Its parameters therefore derive from measured device data.

Each replicated composition cell is summarised by a *parameter card*: the model-free half-enhancement time $t_{1/2,c}$, any identified stretch exponent β_c , the potentiation growth exponent α_c , the measured peak ratio $R_{\text{peak},c}$, and the pulse count $N_{\text{peak},c}$ at that peak. The fading memory is the stretched-exponential (Kohlrausch) retention identified in chapter 4,

$$\lambda_c(\Delta t) = \exp\left[-(\Delta t/\tau_c)^{\beta_c}\right],$$

where $\tau_c = t_{1/2,c}/(\ln 2)^{1/\beta_c}$ is derived so that every card reproduces the primary, model-free cell-median half-time exactly. For cells without an identified stretch exponent, $\beta_c = 1$ and therefore $\tau_c = t_{1/2,c}/\ln 2$. The

write envelope is

$$\frac{G(N)}{G_0} = 1 + (R_{\text{peak},c} - 1) \left(\frac{N}{N_{\text{peak},c}} \right)^{\alpha_c}, \quad 0 \leq N \leq N_{\text{peak},c}.$$

It uses the measured baseline, exponent, peak, and pulse count rather than an unbounded power law. Where a measured train turns over after its peak, the node is capped at the peak: the post-peak recovery dynamics were not measured and are not extrapolated.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt , a single device node updates as

$$\begin{aligned} \ell_n^{(i)} &= \lambda_i x_{n-1}^{(i)}, & p_n^{(i)} &= \text{clip} \left[(\ell_n^{(i)})^{1/\alpha_i} + \frac{\text{clip}(w_i u_n, 0, 1)}{N_{\text{peak},i}}, 0, 1 \right], \\ x_n^{(i)} &= (p_n^{(i)})^{\alpha_i}, & \frac{G_n^{(i)}}{G_0^{(i)}} &= 1 + (R_{\text{peak},i} - 1) x_n^{(i)}, \quad \text{quad} \lambda_i = \exp \left[-(\Delta t / \tau_i)^{\beta_i} \right]. \end{aligned}$$

Here $x_i \in [0, 1]$ is the normalised enhancement, w_i is a non-negative input mask, and one full-scale sample advances the pulse-progress coordinate by at most one effective pulse. That sample-to-pulse mapping is a modelling convention, not a measured circuit timing relation. With no leakage and unit input the recurrence reaches the measured peak after $N_{\text{peak},i}$ updates and cannot exceed it; with no input it follows the measured decay. The internal state is assumed to be read sub-threshold without disturbance, as in the proof-of-concept device of chapter 2. Every read-out is a single *linear* ridge regression on the device states.

Write nonlinearity and fading memory were measured at different write amplitudes (3 V for potentiation and 4 V for depotentiation; table 4.1). Combining them in one node assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. Both measurements also used a single, effectively common inter-pulse cadence, so the model does not resolve input timing below that cadence.

The gold/TMPE negative-polarity corpus of chapter 4 measures integration and relaxation within single runs and finds both interval-dependent potentiation and written-depth-dependent relaxation. Because it uses another host, electrode, polarity, and read protocol, it does not provide parameters for the banks used here. The experiment needed for the lead composition is specified in section 5.11. Per-cell parameter cards and all simulation and read-out settings are tabulated in Appendix D.

5.3 Reservoir Computing and the Common Metric Language

The reservoir principle. Reservoir computing uses a fixed dynamical system whose state is connected to a trained linear read-out [1–3]. A physical reservoir requires fading memory, so its state retains recent input history, and a nonlinear response that maps inputs into features accessible to the read-out. Chapter 4 measures both properties in these devices: stretched-exponential depotentiation supplies fading memory, and compressive potentiation supplies nonlinearity. Memristive and organic implementations have previously used the same combination [4–7]. The behavioural models test whether the measured polymer-electrolyte response supplies that combination.

Architecture. The reservoir is a spatial bank of N *independent* device nodes, each represented by the behavioural model of section 5.2 and driven by a randomly masked copy of the input. With no inter-node recurrence, differences between banks arise from their intrinsic timescales, device-to-device scatter, and input masks rather than engineered connectivity. For Demonstration A, the same single-composition node could instead be deployed as a time-multiplexed delay-feedback reservoir with virtual

nodes along a delay line [8]; the spatial bank is used throughout for consistency.

Common metrics. Linear memory capacity MC_k and its total [9] quantify the past input recoverable at each lag. NARMA-10 [10] provides a nonlinear system-identification benchmark, and the information-processing-capacity framework [11] separates linear and nonlinear contributions (fig. 5.3). Affective tasks are scored by class-balanced macro-F1. Robustness is tested by injecting the measured within-cell timescale spread into the bank (fig. 5.8); no common scatter estimate is assumed for α or β .

5.4 Benchmark Tier: Capacity and System Identification

Reservoir suitability is first evaluated without an application-specific task. Linear memory capacity measures fading memory, while information-processing capacity separates its linear and nonlinear components. The NARMA-10 system-identification task is used only to rank compositions, not to claim absolute performance. All three benchmarks use a random rate-coded input, N device nodes, and a linear ridge read-out.

Statistical units. Random input and bank seeds sample the algorithm’s sensitivity to stochastic initialisation; they are not independent devices, experiments, or subjects. Seed-only benchmark comparisons therefore report means, standard deviations, and sign consistency without inferential p -values. Inferential paired tests are reserved for independent held-out subjects in the physiological-corpus analyses, or for independently fabricated devices in the experimental chapters. The complete

distinction is recorded in Appendix D.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total is the sum over all lags, bounded above by the number of nodes N [9]. Figure 5.1 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09) and a *heterogeneous* bank spanning the replicated composition grid ($\tau \approx 4\text{--}30\text{ s}$). Under the bounded, peak-calibrated update, total capacity changes from 4.19 ± 0.42 to 4.52 ± 0.67 (means $\pm$ SD over ten random input/bank seeds), a ratio of 1.08 ± 0.15 and a paired gain of 0.33 ± 0.60 . The heterogeneous value is higher in seven of ten seed pairs, so the linear-memory benchmark does *not* establish a robust heterogeneity advantage. It does show a different lag profile: single-step recall rises from $MC_1 = 0.52$ to 0.99 , whereas recall at lag 15 falls from 0.11 to 0.05 . The composition spread therefore redistributes recoverable information toward short lags in this architecture rather than increasing total linear capacity decisively.

Composition sweep and the single-node optimum. The same benchmarks, run on single-composition banks, rank the compositions against one another and so test the Demonstration-A claim that one cell suffices for a single-timescale task. Figure 5.2 shows that the lead cell PEO 0.3/0.09 attains the highest single-composition memory capacity (3.76 ± 0.31 , mean $\pm$ SD over ten seeds), while the neighbouring PEO 0.3/0.045 cell is marginally better on the NARMA-10 system-identification benchmark [10] (NRMSE 0.586 ± 0.021 against the lead’s 0.594 ± 0.030). Both optima fall in the low-PEO row that chapter 4 identifies as carrying the longer fading-memory times. This supports the

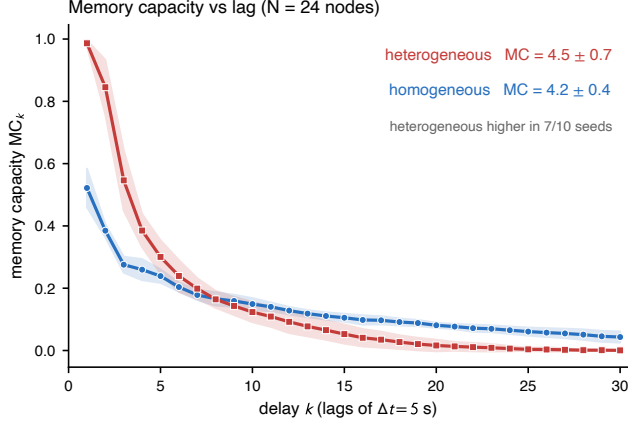


Figure 5.1: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 4\text{--}30$ s). Under the bounded, peak-calibrated node model, total capacity is 4.19 ± 0.42 and 4.52 ± 0.67 , respectively; the heterogeneous value is higher in seven of ten seed pairs. Its main effect is a redistribution toward the shortest lags, not a resolved increase in total linear capacity. Shaded bands are ± 1 SD over ten random input/bank seeds. Seeds quantify algorithmic sensitivity rather than experimental replication. $N = 24$ independent leaky nodes, random uniform input, linear ridge read-out.

narrower Demonstration-A statement that the lead composition is a reasonable single-timescale operating point; the small differences between the two low-PEO cells are algorithmic rankings, not experimental evidence of a unique optimum. The NARMA-10 numbers carry only *relative* weight: the absolute error is high ($\approx 0.6\text{--}0.9$) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir.

Comparison with randomly distributed timescales. A generic echo-state bank of the same size tests whether the measured timescales are individually important or whether their range is sufficient. A random bank spanning approximately the same range ($\tau = 3\text{--}30$ s) has a memory capacity of 4.97 ± 0.78 , compared with 4.52 ± 0.67 for the device bank

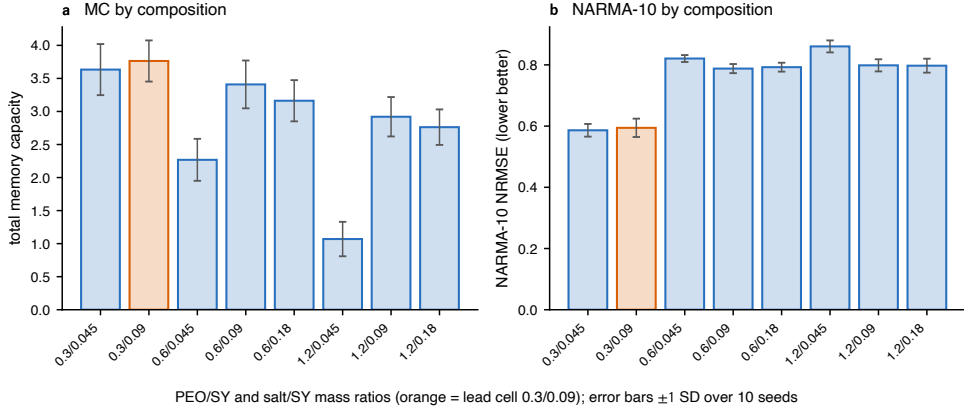


Figure 5.2: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The lead cell PEO 0.3/0.09 (orange) attains the highest memory capacity; the neighbouring 0.3/0.045 cell is marginally better on NARMA-10. Both optima fall in the low-PEO row anticipated by chapter 4. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

(ten matched seeds; the device is higher in four). Expanding the random range to 0.5–60 s raises the capacity to 5.56 ± 0.74 (the device is higher in two). The measured values are therefore not uniquely optimal; this control attributes the available linear memory to the range of fading times rather than to their exact distribution. The broader-range result is only a modelling prompt: the corresponding slow devices have not been measured.

Linear and nonlinear capacity. Memory capacity counts only linearly recoverable information. The information-processing-capacity framework of Dambre et al. decomposes total capacity into orthogonal polynomial degrees: degree-one terms give linear capacity, while degree-two terms measure nonlinear self and cross-lag transformations; the total is bounded by the node count N [11]. Figure 5.3 reports this decomposition. The

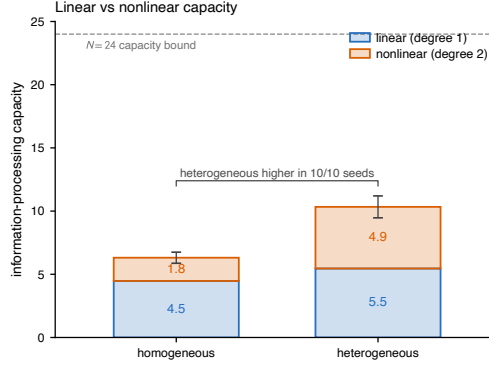


Figure 5.3: Information-processing capacity [11] split into linear (degree-one) and nonlinear (degree-two) parts for homogeneous and heterogeneous banks ($N = 24$, ten seeds). Both banks have nonzero degree-two capacity under the bounded node model. Mean total capacity changes from 6.31 ± 0.44 to 10.33 ± 0.87 ; the nonlinear component changes from 1.84 to 4.88 and is higher in all ten seed pairs. The dashed line is the $N = 24$ upper bound; seeds quantify algorithmic sensitivity rather than experimental replication.

homogeneous bank has total capacity 6.31 ± 0.44 , comprising 4.47 linear and 1.84 nonlinear capacity. The heterogeneous bank reaches 10.33 ± 0.87 , comprising 5.46 linear and 4.88 nonlinear capacity; its nonlinear increment of 3.04 is positive in all ten seed pairs. Of the heterogeneous nonlinear capacity, 2.67 is from self terms and 2.21 from cross-lag products. This is a stronger algorithmic result than the total linear-memory comparison, but still a sensitivity result for the composed behavioural model rather than an experimental population estimate.

5.5 Affective Computing

Affective computing uses physiological signals to infer emotional and physiological state [12]. The device timescales, from a few seconds to tens of seconds, overlap the phasic electrodermal, respiration, and heart-

Table 5.1: Characteristic timescales of the peripheral affective signals used in this chapter, and the composition cells whose fading-memory window covers them [13–15]. The grid of chapter 4 spans the full range, with the lead PEO 0.3 cells serving the slow LF-HRV band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	$\sim 1\text{--}10\text{ s}$	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	$\sim 3\text{--}5\text{ s}$	mid PEO
HRV, HF band	$\sim 2.5\text{--}7\text{ s}$	mid PEO
HRV, LF band	$\sim 7\text{--}25\text{ s}$	PEO 0.3 ($t_{1/2} \approx 16\text{--}19\text{ s}$)
Tonic SCL	minutes	slowest / drive-boosted

rate-variability bands without time rescaling (table 5.1). These signals also contain several timescales at once: phasic electrodermal responses, respiration and heart-rate-variability rhythms over seconds, and tonic levels drifting over minutes [13–15]. Figure 5.4 places the measured relaxation time of each composition on the same axis as these physiological bands and shows the minutes-scale range not reached by composition alone.

Extending the model to tonic timescales. Raising the write amplitude lengthens fading memory in the measurements of chapter 4. Adding modelled nodes at $\tau = 120$ and 300 s raises the random-input long-lag capacity beyond 75 s from 0.20 to 0.91 and total capacity from 4.52 to 5.03 ; both changes are positive in all ten seed pairs (Appendix D). On the physiological streams, reconstruction at delays up to 120 s is reassessed in section 5.6. This extension is explicitly hypothetical because no minutes-scale cell was measured and the drive-amplitude data do not identify these two time constants. A matched drive-amplitude experiment at these timescales is required to test it.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [16]: fifteen subjects wearing a chest

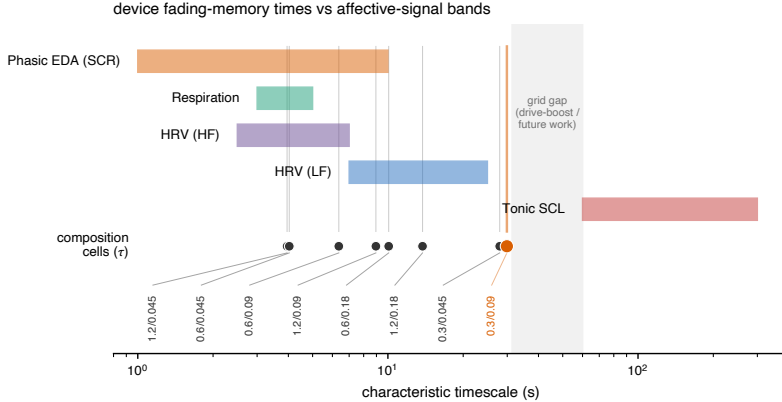


Figure 5.4: Each replicated PEO/LiOTf composition cell is placed at its derived relaxation time τ (markers, lower track) on the same logarithmic axis as characteristic timescale bands of peripheral affective signals (coloured bars; table 5.1). The grid ($\tau \approx 4\text{--}30\text{ s}$, anchored to measured $t_{1/2}$) overlaps the phasic-EDA, respiration, and HRV bands, with the lead cell PEO 0.3/0.09 (orange) near the slow LF-HRV band. Composition alone does not reach the tonic, minutes-scale band (shaded gap); testing that range would require a measured slower cell or matched drive-amplitude experiments.

sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways: as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.6), and as a downstream affective-classification task (section 5.7).

5.6 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.4 show that a spread of device timescales broadens memory on a random input. This section tests the same prop-

erty on the multi-timescale physiological signals introduced in section 5.5, isolating temporal-context encoding before turning to affect labels (section 5.7).

The reservoir is run continuously over each WESAD session at $\Delta t = 1$ s on four chest channels (electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate), and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [13, 14]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Preprocessing is causal. ECG filtering and RR-interval assignment use only completed beats; resampling uses the nearest past label. Each channel is mapped to $(0, 1)$ by a monotone transform fitted to the first 60 s only: the calibration-prefix 5th and 95th percentiles map to 0.25 and 0.75, without hard clipping later values. Thus neither later held-out values nor affect labels set the scale. The reported out-of-sample R^2 uses the training-subject mean as its deployable null predictor, clips reconstructed normalised physiology to $[0, 1]$, and weights each held-out subject equally.

Table 5.2 reports the held-out reconstruction quality. A memoryless bank gives $R^2 = 0.636$, compared with 0.646 for instantaneous input. A single-timescale memory raises R^2 to 0.679–0.692, and the heterogeneous composition bank reaches 0.701: +0.055 over instantaneous input and +0.010 over the stronger same-size homogeneous control. The heterogeneous-minus-fast difference is positive in 14/15 subjects (Holm-adjusted $p = 2.4 \times 10^{-4}$); heterogeneous-minus-slow is positive in 11/15 ($p_{\text{Holm}} = 0.030$). The fast homogeneous bank is strongest at the shortest delays and the slow homogeneous bank at the longest; heterogeneity gives the highest overall score, but the margin is modest (fig. 5.5).

Table 5.2: Subject-mean leave-one-subject-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). The null prediction is the training-subject mean. Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. Across the full context vector, the heterogeneous bank exceeds the stronger homogeneous control by +0.010 R^2 (11/15 subjects; Holm-adjusted $p = 0.030$).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.646	0.699	0.647	0.592
Memoryless	0.636	0.691	0.637	0.580
Homogeneous, fast	0.679	0.777	0.655	0.594
Homogeneous, slow	0.692	0.714	0.702	0.664
Heterogeneous	0.701	0.767	0.696	0.638

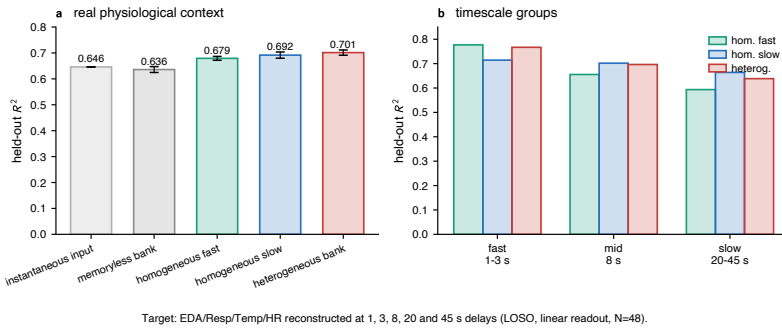


Figure 5.5: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and causally derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s (leave-one-subject-out, $N = 48$). The heterogeneous bank gives the highest overall subject-mean R^2 (0.701), ahead of the slow homogeneous control (0.692) and instantaneous input (0.646). The zero-based axes avoid magnifying these modest differences.

5.7 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with a leave-one-subject-out linear read-out. The reservoir and the 15 s causal prediction smoother run on the complete

Table 5.3: Matched comparison on three-class WESAD chest signals. Both rows report class-balanced macro-F1 under leave-one-subject-out validation over the same fifteen subjects. The models and temporal formulations still differ, so the table provides a scale reference rather than a superiority claim. Accuracy-only and differently split studies are omitted.

Study	Features and read-out	Validation	macro-F1
Schmidt et al. (2018) [16]	engineered window features, AdaBoost/RF	15-subject LOSO	≈ 0.72
This work	streaming device-model states, linear ridge read-out	15-subject LOSO	0.524

physical timeline; samples with undefined affect labels are excluded only from fitting and scoring. This prevents the smoother from joining labelled blocks separated by long protocol gaps. The study that introduced WESAD reports three-class chest macro-F1 ≈ 0.72 with engineered window features and conventional classifiers under leave-one-subject-out validation [16]; the present streaming linear read-out reaches 0.524 averaged over bank draws (table 5.3). The formulations differ, so this is a scale reference and not a like-for-like ranking.

Demonstration A: window-level electrodermal classification. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches leave-one-subject-out macro-F1 0.669 (fig. 5.6a). A static baseline using the mean, standard deviation, last value, and slope of the same signal reaches 0.714. The reservoir does not improve this 60 s-window task.

Demonstration B: streaming three-class tracking. The reservoir runs continuously over each session, and baseline, stress, or amusement is read out at every labelled step. Table 5.4 decomposes the result into four matched-size banks. A majority-class predictor reaches macro-F1 0.231

Table 5.4: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means $\pm$ SD over ten matched bank seeds). The homogeneous fading-memory bank is strongest; adding the measured timescale spread does not improve the sustained-label task.

Bank	macro-F1	increment
Instantaneous (no memory)	0.482	—
Memoryless (dimensionality only)	0.513 ± 0.040	+0.031
Homogeneous (memory, single τ)	0.545 ± 0.037	+0.032
Heterogeneous (memory, τ spread)	0.524 ± 0.032	−0.021

and a stratified-random predictor 0.333. Instantaneous input reaches 0.482; the memoryless bank reaches 0.513 ± 0.040 ; and the homogeneous fading-memory bank reaches 0.545 ± 0.037 . The heterogeneous bank reaches 0.524 ± 0.032 , a +0.042 gain over instantaneous input but -0.021 ± 0.024 relative to homogeneous, with heterogeneity higher in only one of ten matched seed pairs. At the representative seed, amusement remains the weakest class (F1 0.35, versus 0.82 for baseline and 0.74 for stress). The corrected experiment therefore supports a memory benefit relative to an instantaneous read, but not a timescale-diversity benefit for these sustained labels.

Why heterogeneity does not improve the label scores. The affect labels are slowly varying sustained states, and their discriminative memory is dominated by the slow tonic band already covered by the lead composition. A wider timescale distribution is modestly useful when the target explicitly requires several lags (table 5.2), but not here. Under this protocol, fading memory helps classification; replacing the homogeneous bank with the measured heterogeneous distribution slightly lowers the seed-mean score.

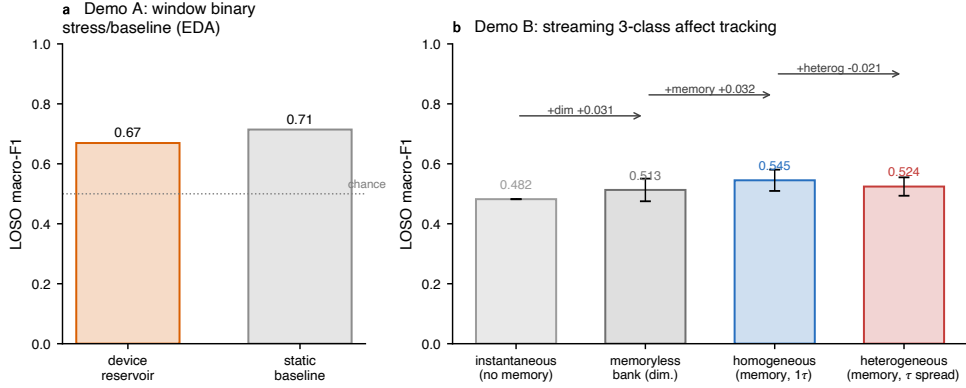


Figure 5.6: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Window-level binary stress-versus-baseline from EDA: the device model scores 0.669, below the static-feature control (0.714). (b) Streaming three-class tracking. Fading memory improves on instantaneous input, but the homogeneous bank (0.545 ± 0.037) exceeds the heterogeneous bank (0.524 ± 0.032). Error bars are ± 1 SD over ten bank seeds; both bar-chart axes begin at zero.

5.8 Continuous Monitoring Under Deployment Conditions

The sustained WESAD states can often be classified from the instantaneous signal level, as shown by the static-feature baseline. Wearable measurements also contain sensor noise and motion artefacts, particularly in wrist-worn electrodermal and photoplethysmographic signals. Fading memory implements temporal low-pass filtering and may therefore reduce sensitivity to transient artefacts. The tests below add synthetic noise, measured-motion gating, and an independent sensor corpus.

Figure 5.7 tests a continuous binary stress detector (stress versus baseline plus amusement, read out at every step, leave-one-subject-out, with the same linear read-out used throughout). On clean streams, macro-F1 ranges from 0.781 to 0.800. At $\sigma = 0.4$, instantaneous and memo-

ryless reads fall to 0.641 and 0.649, while the homogeneous and heterogeneous memory banks retain 0.807 and 0.775. The heterogeneous-minus-instantaneous subject difference is +0.125 (12/15 positive, paired Wilcoxon $p = 0.030$). Clean false-alarm rates are similar (instantaneous 0.094, heterogeneous 0.090 seed-mean). Transition-window F1 changes from 0.832 to 0.848, but the subject difference is not resolved (6/15 positive, $p = 0.33$); all banks have zero median onset latency under the five-second stable-call definition. Temporal smoothing is useful under corruption, not for earlier detection in this protocol.

Accelerometer-gated motion artefact. Because the preceding corruption is synthetic, a second test gates its amplitude with the measured chest accelerometer. At the strongest gating, the instantaneous read falls to 0.702, the heterogeneous bank retains 0.791, and the EMA reaches 0.810. The heterogeneous-minus-instantaneous subject difference (+0.087) is not resolved ($p = 0.084$), whereas EMA-minus-instantaneous is (+0.112, $p = 0.035$; Appendix D). The data-driven control therefore supports conventional temporal filtering under measured-motion corruption, but not a separate device-model advantage.

Classical-filter control. A single exponential moving average reaches macro-F1 0.804 at $\sigma = 0.4$, compared with 0.641 for instantaneous input; its paired subject gain is +0.171 ($p = 0.022$). A linear multi- τ EMA bank reaches 0.811, with no resolved gain over the single EMA (+0.006, $p = 0.20$). The heterogeneous device model reaches 0.775 and is lower than the EMA bank by -0.052 ($p = 0.030$; Appendix D). Robustness is therefore attributable to temporal low-pass filtering, and the linear digital control is superior on this test. The proposed material role is passive filtering, not an accuracy advantage over that control.

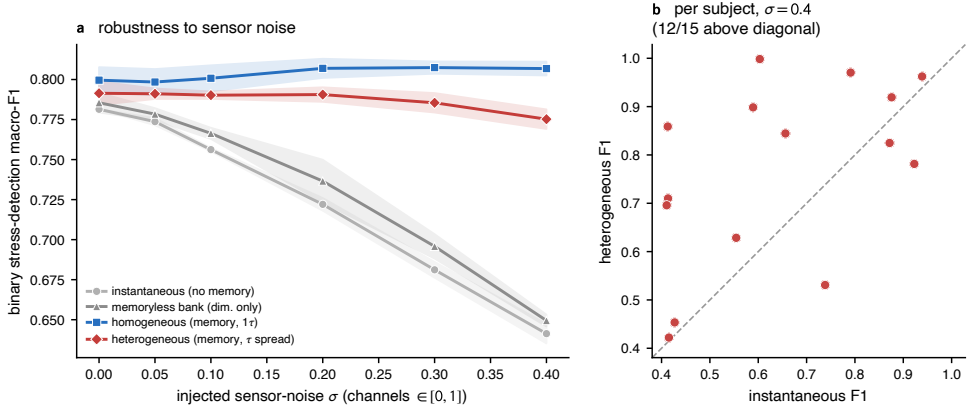


Figure 5.7: Continuous stress monitoring (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against injected corruption. At $\sigma = 0.4$, instantaneous input falls to 0.641, while homogeneous and heterogeneous memory banks retain 0.807 and 0.775; bands are ± 1 SD over seeds. (b) The heterogeneous bank exceeds instantaneous input in 12/15 subjects ($p = 0.030$). The homogeneous bank’s higher score shows that timescale spread is not required.

Cross-corpus replication. The robustness test was repeated on the PhysioNet Non-EEG corpus of Birjandtalab et al. [17]: twenty subjects wearing an Affectiva Q wrist sensor for electrodermal activity and skin temperature, with pulse-oximeter heart rate. Cognitive and emotional stress are compared with relaxation; physical exertion is excluded. On clean streams, seed-mean heterogeneous and instantaneous scores are 0.644 and 0.639; the representative-seed subject difference is not resolved ($p = 0.39$). At $\sigma = 0.4$, they are 0.629 and 0.451: a +0.177 gain, positive in all twenty subjects ($p = 1.9 \times 10^{-6}$; table 5.5). This independent corpus reproduces the noise-response ordering, while confirming that the clean-stream advantage is absent.

Table 5.5: Noise-response comparison on the independent PhysioNet Non-EEG stress corpus [17] (twenty subjects; leave-one-subject-out, linear read-out). At $\sigma = 0.4$, the heterogeneous-minus-instantaneous difference is $+0.177$, positive in 20/20 subjects ($p = 1.9 \times 10^{-6}$). Clean-stream scores are comparable.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.639	0.451
Homogeneous (memory)	0.652	0.630
Heterogeneous (memory)	0.644	0.629

5.9 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are addressed: whether the advantage survives a consumer-grade sensor, and what can be said about an operating budget. The first is tested with the WESAD wrist recordings. For the second, only a reference calculation is possible because the simulated PEO nodes were not measured as a wired bank.

Sensor site. WESAD records each session with both a chest strap and an Empatica E4 wrist band. The wrist band has fewer and noisier channels: skin conductance, skin temperature, and photoplethysmography-derived heart rate, without respiration. On clean wrist data (table 5.6), instantaneous and heterogeneous scores are 0.724 and 0.736. At $\sigma = 0.4$ they are 0.615 and 0.732; the paired subject gain is $+0.106$ (11/15 positive, $p = 0.035$). The corresponding chest scores are 0.641 and 0.775. Clean wrist false-alarm rates are not improved (instantaneous 0.112, heterogeneous 0.115), so the supported benefit is robustness under injected corruption rather than fewer clean-stream alarms.

Table 5.6: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out; seed means). Under injected corruption, the temporal bank exceeds instantaneous input at both sites; σ follows fig. 5.7.

Detector	Chest		Wrist E4	
	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.781	0.641	0.724	0.615
Memory reservoir (het)	0.791	0.775	0.736	0.732

Operating-budget reference. The parameter count is exact for the simulated architecture: the device dynamics are fixed, and the fitted $(N+1) \times K$ read-out contains 75 weights for $N = 24$, $K = 3$, or 147 for $N = 48$. The read-out also requires NK multiply-accumulates per step. Energy is less certain. Recalculation of one proof-of-concept *Super Yellow/Hybrane*/LiOTf junction gives $0.084 \mu\text{J}$ – $1.55 \mu\text{J}$ across its positive write sequence (median $1.08 \mu\text{J}$). Multiplying that median by 24 nodes gives a reference core cost of $25.9 \mu\text{J}$ per update, or $25.9 \mu\text{W}$ at one update per second; the corresponding 48-node value is $51.8 \mu\text{W}$.

These products are not power predictions for the proposed reservoir. They transfer a 1 V result from one large-area *Hybrane* junction to PEO models identified at 3 V–4 V, without measured bank wiring, read energy, or peripheral circuits. They exclude the sensor analogue front-end, analogue-to-digital conversion, addressing, and control, and are neither an upper nor a lower bound. The device response remains faster than the sub-hertz physiological signals considered here, and its relaxation would perform the temporal filtering during the write/read cycle. The assumptions are tabulated in Appendix D.

5.10 Matching Device and Task Timescales

Performance depends on the overlap between device and task timescales (fig. 5.4). In the single-composition sweep, the lead cell gives the highest memory capacity for the seconds-scale benchmark (fig. 5.2). For the multi-lag physiological task, no single cell covers the full delay range, and the heterogeneous bank gives the highest single-bank score (table 5.2).

Effect of timescale breadth. Table 5.7 separates fading memory from timescale breadth. The bounded benchmark does not resolve a total linear-memory advantage: the heterogeneous-minus-homogeneous difference is $+0.33 \pm 0.60$, positive in 7/10 seeds. It does raise nonlinear information-processing capacity and change the lag profile. On real physiology, heterogeneity gives a modest $+0.010$ context-reconstruction gain over the stronger homogeneous control, but loses 0.021 macro-F1 on sustained three-class labels. Under corruption, the homogeneous bank is also stronger. The task, not diversity alone, determines the useful distribution.

When one slow timescale dominates, as in the sustained stress labels and stress-monitoring task, the best-matched homogeneous bank is better by about 0.01 – 0.03 macro-F1. The available variation comes from composition and device-to-device scatter; chemistry and drive amplitude provide additional, sample-limited axes in chapter 4.

Operating point and integration. The models apply in the measured sub-kilohertz regime, which covers the physiological signals used here but not fast logic. The assumed sub-threshold, state-preserving read sets the read-voltage and read-disturb limits for a circuit implementation. Figure 5.8 varies timescale-only jitter. Its primary value, $\sigma(\ln t_{1/2}) \approx 0.264$, is the median within-cell substrate-level spread on the screened

Table 5.7: Comparison of instantaneous, homogeneous, and heterogeneous banks across the affective tasks in this chapter. Temporal banks show their largest gains under corruption. Heterogeneity gives the highest physiological-context score, whereas the homogeneous bank is strongest on most classification rows. Bold marks the highest score per row.

Affective task	Metric	Instantaneous	Homogeneous	Heterogeneous
Physiological context (multi- τ)	R^2	0.646	0.692	0.701
Three-class affect tracking	macro-F1	0.482	0.545	0.524
Stress monitoring, clean	macro-F1	0.781	0.800	0.791
Stress monitoring, $\sigma=0.4$	macro-F1	0.641	0.807	0.775
Wrist stress, clean	macro-F1	0.724	0.730	0.736
Wrist stress, $\sigma=0.4$	macro-F1	0.615	0.738	0.732

replicated grid (Appendix D). The observed per-cell estimates span 0.089–1.018, so the wider sweep is a sensitivity analysis rather than a fitted uncertainty distribution. The legacy single-exponential fit gives a larger $\sigma(\ln \tau) \approx 0.85$, retained only as an upper stress-test value. No measured common jitter is applied to α or β . At zero timescale scatter, copies of one card are more redundant; the heterogeneous bank still differs through composition-dependent card parameters.

5.11 Limitations and Scope Conditions

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench reservoir. Every figure caption names the data behind it. A wired hardware reservoir remains future work.

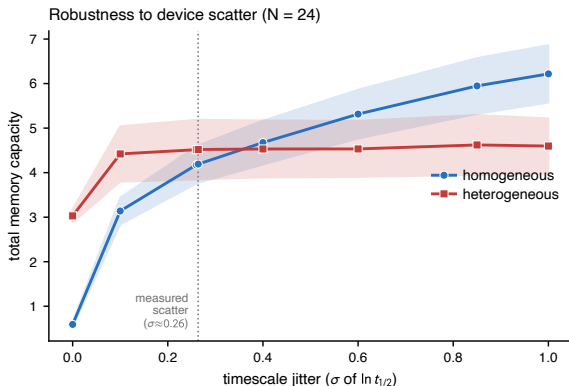


Figure 5.8: Total memory capacity versus injected log-normal timescale jitter for homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The dotted line is the primary screened, substrate-level estimate $\sigma(\ln t_{1/2}) = 0.264$ (Appendix D). Only the timescale is perturbed; α and β retain their measured cell-card values. Larger jitter values, including the legacy fit-based value near 0.85, are stress tests. The curves describe algorithmic sensitivity and do not establish variability as beneficial.

The composition assumption. As detailed in section 5.2, write non-linearity and fading memory were measured separately, at different write amplitudes and one inter-pulse cadence. Combining them assumes that the two regimes are compatible; even the direction and magnitude of the omitted coupling are uncertain for the silver/PEO system. In a sensitivity analysis, each node’s retention was rescaled with its write exponent over $\kappa = -0.5$ to $+0.5$. The mean heterogeneous-to-homogeneous capacity ratio remains small, 1.04–1.09 (Appendix D). The gold/TMPE negative-polarity measurements provide an external directional reference, $\rho = +0.46$ between write depth and decay half-time across 28 single-run traces. The nominal cards already have a stronger across-card rank association ($\rho \approx +0.57$), so the central external estimate maps to $\kappa^* = 0$ and its upper confidence limit to about $+0.31$; the capacity ratio is 1.08 at both points. Because host, electrode, polarity, and protocol differ, this is a sensitivity anchor rather than validation of the composed silver model.

A matched-amplitude experiment on the lead composition that varies pulse count and inter-pulse interval together is required.

Operating regime and read-out. The model is identified only in the measured sub-kilohertz regime; fast temporal tasks are outside its scope. Every read-out is a single ridge regression because a nonlinear read-out could supply the task nonlinearity independently of the reservoir. The read is assumed to be sub-threshold and state-preserving.

Heterogeneity axes. The quantitative comparisons span the replicated PEO/LiOTf composition grid; the single recovered PEO 0.15 substrate is excluded. Chemistry and drive amplitude enter only as secondary controls because their matched cells contain at most two substrates (chapter 4); Demonstration B is a proof of principle for composition heterogeneity, not a powered comparison across chemistries. The cards index composition, but thickness and other fabrication variables are not fully separable from that design. The available covariate checks do not explain the composition association, yet they cannot prove composition is the sole cause (fig. 4.1, section 4.2).

Corpora, protocol, and negative controls. The affective results use WESAD as the main corpus, and noise robustness is replicated on the PhysioNet Non-EEG dataset (section 5.8). A broader multi-corpus study remains future work. In WESAD, the experimental blocks follow a fixed protocol rather than a counterbalanced order. Affect label, elapsed time, and slow session drift are therefore partly entangled: causal prefix calibration prevents future-sample leakage, but it cannot establish that a classifier is specific to affect rather than protocol position. The sustained block labels also do not provide independent ground truth for physiological onset, so the transition and latency analyses quantify agreement with

protocol boundaries, not the onset of an internal state.

The strongest robustness results use synthetic, channel-wise corruption and should be read as controlled stress tests. The measured-motion gate is a more realistic but weaker control and does not resolve a heterogeneous-over-instantaneous advantage. Two further comparisons do not show a device-specific improvement: timescale heterogeneity lowers the seed-mean final-label score relative to the slow homogeneous control (section 5.7), and the nonlinear device encoding is inferior to a linear multi- τ EMA under strong synthetic corruption (Appendix D). The corrected random-input benchmark likewise does not resolve a total linear-memory advantage, although it does show higher nonlinear capacity and a shifted lag profile.

5.12 Summary

Behavioural models fitted to the measurements of chapter 4 support temporal-computing simulations with a linear read-out. Once the node update is bounded by the measured potentiation peak and pulse count, the heterogeneous bank changes total linear memory only from 4.19 to 4.52 (7/10 seeds), but increases mean information-processing capacity from 6.31 to 10.33, chiefly through nonlinear degree-two terms. On WESAD, a lead-cell reservoir scores 0.669 on windowed binary EDA, below a static control (0.714). Streaming three-class macro-F1 is 0.524 ± 0.032 for the heterogeneous bank and 0.545 ± 0.037 for homogeneous memory. Under synthetic corruption, the heterogeneous bank retains 0.775 while instantaneous input falls to 0.641, but a linear EMA bank reaches 0.811 and significantly exceeds the device model.

The additional value of a timescale distribution depends on the task and metric. It is not resolved for total random-input linear memory or sustained WESAD labels. It is large for nonlinear capacity and modest

but subject-resolved for multi-lag physiological reconstruction ($+0.010 R^2$ over the slow homogeneous control). Noise robustness replicates on the independent Non-EEG corpus, but represents generic temporal filtering rather than a device-specific accuracy advantage.

The next experimental steps are a matched-amplitude pulse protocol that varies pulse count and cadence, a replicated study of chemistry and drive-amplitude heterogeneity, and a wired device bank that tests the behavioural-model predictions in hardware.

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